

AKSHAY V

24BECS174

AIML PROGRAM 4

Programiz
Python Online Compiler

main.py

Visualize

Run

```
1 def find_s(training_data):
2     # Initialize the specific hypothesis with the first positive instance
3     hypothesis = None
4
5     for row in training_data:
6         if row[-1] == 'Yes': # Target concept is positive
7             hypothesis = row[:-1].copy()
8             break
9
10    if hypothesis is None:
11        return "No positive instances found."
12
13    # Update hypothesis based on remaining positive instances
14    for row in training_data:
15        if row[-1] == 'Yes':
16            for i in range(len(hypothesis)):
17                if row[i] != hypothesis[i]:
18                    hypothesis[i] = '?'
19
20    return hypothesis
21
22 # Example Usage
23 if __name__ == "__main__":
24
25     # Columns:
26     # Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
27
28     dataset = [
29         ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
30         ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
31         ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
32         ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
33     ]
34
35     most_specific_hypothesis = find_s(dataset)
36
37     print(
38         "Most Specific Hypothesis found by Find-S:",
39         most_specific_hypothesis
```

Output

Most Specific Hypothesis found by Find-S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

=== Code Execution Successful ===